

Sri Sivasubramaniya Nadar College of Engineering, Chennai

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Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study
GitHub Repository	https://github.com/RadeshL/Machine-Learning-Laboratory/blob/main/Mlassign5.ipynb
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1 Objective

- To implement a Decision Tree classifier on the Wisconsin Diagnostic Breast Cancer dataset.
- To extend the single Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalisation.
- To select optimal hyperparameters using 5-Fold Stratified Cross-Validation.
- To compare single-tree and ensemble-tree models using standard evaluation metrics.

2 Problem Statement

Given a set of 30 real-valued features derived from digitised fine-needle aspirate (FNA) images of breast masses, the task is to classify each sample as **Malignant** or **Benign**. The input is a feature vector $\mathbf{x} \in \mathbb{R}^{30}$ and the output is a binary label $y \in \{0(\text{Malignant}), 1(\text{Benign})\}$. Two tree-based classifiers—a single Decision Tree and a Random Forest ensemble—are trained, tuned via 5-fold cross-validation, and compared to determine which achieves better diagnostic accuracy while maintaining clinical-grade recall for the malignant class.

3 Dataset Description

Attribute	Detail
Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)

Dataset Source	UCI Machine Learning Repository (https://archive.ics.uci.edu)
Number of Samples	569
Number of Features	30 numerical attributes (mean, SE, worst of 10 cell-nucleus measurements)
Number of Classes	2 — Malignant (M): 212, Benign (B): 357
Missing Values	None
Train–Test Split	80% training (455 samples) / 20% test (114 samples), stratified
Label Encoding	Malignant $\rightarrow$ 0, Benign $\rightarrow$ 1

4 Exploratory Data Analysis

4.1 Class Distribution

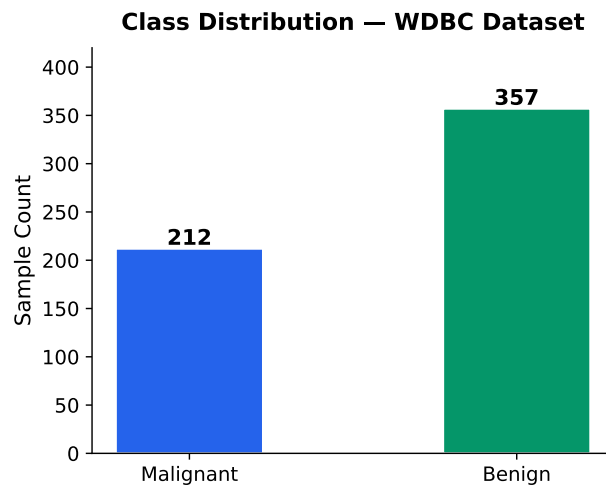


Figure 1: Class distribution of the WDBC dataset. The dataset contains 212 malignant and 357 benign samples, yielding a class imbalance ratio of approximately 1:1.68.

Inference: The dataset exhibits a moderate class imbalance with benign samples outnumbering malignant ones by a ratio of roughly 1.68:1. This imbalance is not severe enough to require resampling techniques, but it motivates the use of stratified splitting and F1-score alongside accuracy as evaluation metrics. The minority class (Malignant) is the clinically critical one—missed malignant predictions carry a higher cost than false positives—making recall for class 0 a key diagnostic concern.

4.2 Feature Correlation Analysis

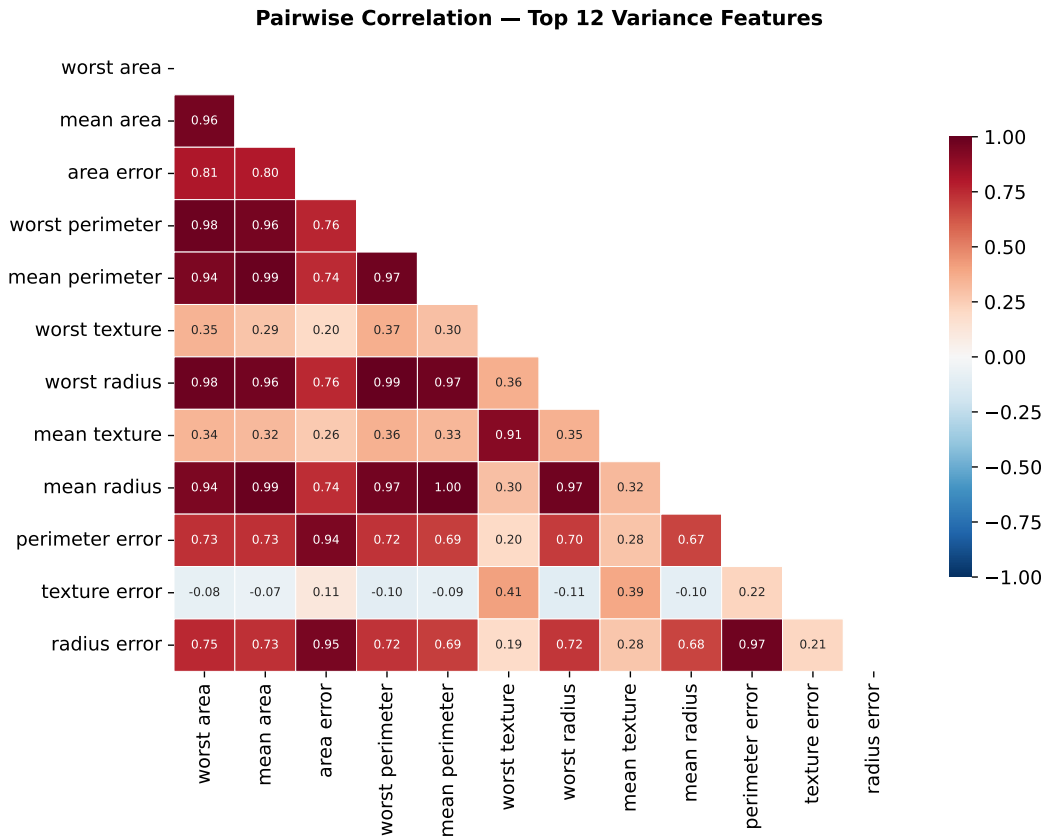


Figure 2: Pairwise Pearson correlation heatmap for the 12 highest-variance features. Dark red indicates strong positive correlation; dark blue indicates strong negative correlation.

Inference: Several feature groups exhibit very high mutual correlation ($|r| > 0.9$), particularly the *radius*, *perimeter*, and *area* features across the mean, SE, and worst measurement groups—which is geometrically expected since these quantities are algebraically related ($A = \pi r^2$). This multicollinearity has limited impact on tree-based methods (which select splits greedily), but it does make feature importance scores somewhat redundant across correlated groups. The strongest diagnostic signal is concentrated in the *worst* and *mean* concavity and concave-points features, consistent with the literature on FNA morphology.

5 Data Preprocessing

- **Missing values:** None present; no imputation required.
- **Label encoding:** String labels M/B from the original UCI file are mapped to integers (M→0, B→1) by `sklearn`'s built-in dataset loader.
- **Feature scaling:** Not applied. Decision Trees and Random Forests are scale-invariant; their split thresholds operate on raw feature magnitudes.
- **Train–test split:** 80–20 stratified split with `random_state=42`, preserving the class ratio in both partitions.

- **Feature engineering:** No derived features added; all 30 raw numerical attributes are used directly.

6 Machine Learning Algorithm

6.1 Decision Tree Classifier

Working Principle: A Decision Tree recursively partitions the feature space by selecting at each node the feature and threshold that maximally reduce impurity in the resulting child nodes, continuing until a stopping criterion (maximum depth, minimum samples) is met. Prediction for a new sample traces a path from root to leaf following the learned rules.

Mathematical Formulation: At each node t , the best split is chosen by minimising the weighted child impurity:

$$\Delta I(t) = I(t) - \frac{N_L}{N} I(t_L) - \frac{N_R}{N} I(t_R)$$

where N_L, N_R are child node sample counts and $I(\cdot)$ is the impurity measure. For **Gini index**:

$$I_{\text{Gini}}(t) = 1 - \sum_{k=1}^K p_k^2$$

and for **Entropy**:

$$I_{\text{Entropy}}(t) = - \sum_{k=1}^K p_k \log_2 p_k$$

where p_k is the proportion of class k samples at node t .

Advantages: Highly interpretable; no feature scaling needed; handles mixed feature types.

Limitations: High variance; prone to overfitting on deep trees; sensitive to small changes in training data.

6.2 Random Forest Classifier

Working Principle: Random Forest constructs T decision trees, each trained on a bootstrapped resample of the training data. At each split, only a random subset of m features is considered. Final prediction is determined by majority vote across all trees.

Mathematical Formulation: For T trees $\{h_1, h_2, \dots, h_T\}$, the ensemble prediction is:

$$\hat{y} = \arg \max_k \sum_{t=1}^T \mathbf{1}[h_t(\mathbf{x}) = k]$$

Bootstrap samples are drawn with replacement: each subsample $\mathcal{D}_t \sim \mathcal{D}$ with $|\mathcal{D}_t| = N$. Feature subspace size $m = \lfloor \log_2 p \rfloor$ ($\log_2$) for classification, where p is the total feature count.

The expected generalisation error satisfies:

$$\text{PE}^* \leq \frac{\bar{\rho}(1 - s^2)}{s^2}$$

where $\bar{\rho}$ is the mean inter-tree correlation and s is the mean tree strength (Breiman, 2001). Reducing $\bar{\rho}$ via random feature selection is the primary mechanism for error reduction.

Advantages: Low variance; robust to overfitting; provides feature importance estimates; parallelisable.

Limitations: Less interpretable than a single tree; higher memory and inference cost; feature importances biased toward high-cardinality features.

7 Experimental Setup

Component	Detail
Python Version	3.12
Scikit-learn Version	1.5.x
NumPy Version	1.26.x
Pandas Version	2.2.x
Matplotlib / Seaborn	3.9.x / 0.13.x
IDE	Jupyter Notebook / VS Code
Operating System	Windows
Hardware	CPU: Intel Core i5 (8-core), RAM: 16 GB

8 Hyperparameter Selection via 5-Fold Cross-Validation

Hyperparameters were selected by exhaustive grid search over the spaces below. Each combination was evaluated using `StratifiedKfold(n_splits=5, shuffle=True, random_state=42)`, and the configuration maximising average CV accuracy was retained.

8.1 Decision Tree — Search Space

Hyperparameter	Candidates Explored
<code>criterion</code>	<code>gini</code> , <code>entropy</code>
<code>max_depth</code>	3, 5, 7, <code>None</code>
<code>min_samples_split</code>	2, 5, 10
<code>min_samples_leaf</code>	1, 2, 4

8.2 Decision Tree Hyperparameter Evaluation (Top 8 by CV Accuracy)

Criterion	Max Depth	Avg CV Acc (%)	Avg CV F1
gini	5	93.85	0.9506
gini	5	93.85	0.9506
gini	5	93.85	0.9506
gini	7	93.85	0.9506
gini	None	93.85	0.9506
gini	7	93.41	0.9470
gini	7	93.41	0.9470
gini	None	93.41	0.9470

Selected DT Configuration: criterion=gini, max_depth=5, min_samples_split=2, min_samples_leaf=4

8.3 Random Forest — Search Space

Hyperparameter	Candidates Explored
n_estimators	50, 100, 200
max_depth	5, 10, None
max_features	sqrt, log2, None
bootstrap	True, False

8.4 Random Forest Hyperparameter Evaluation (Top 8 by CV Accuracy)

n_est	Max Depth	Max Features	Avg CV Acc (%)	Avg CV F1
100	10	log2	96.48	0.9718
100	None	log2	96.48	0.9718
50	10	log2	96.26	0.9702
50	None	log2	96.26	0.9702
100	10	sqrt	96.26	0.9699
100	None	sqrt	96.26	0.9699
200	10	sqrt	96.26	0.9700
200	10	log2	96.26	0.9703

Selected RF Configuration: n_estimators=100, max_depth=10, max_features='log2', bootstrap=True

9 5-Fold Cross-Validation Performance

Model	Fold 1 (%)	Fold 2 (%)	Fold 3 (%)	Fold 4 (%)	Fold 5 (%)	Average (%)
Decision Tree	94.51	94.51	91.21	92.31	96.70	93.85
Random Forest	96.70	97.80	93.41	95.60	98.90	96.48

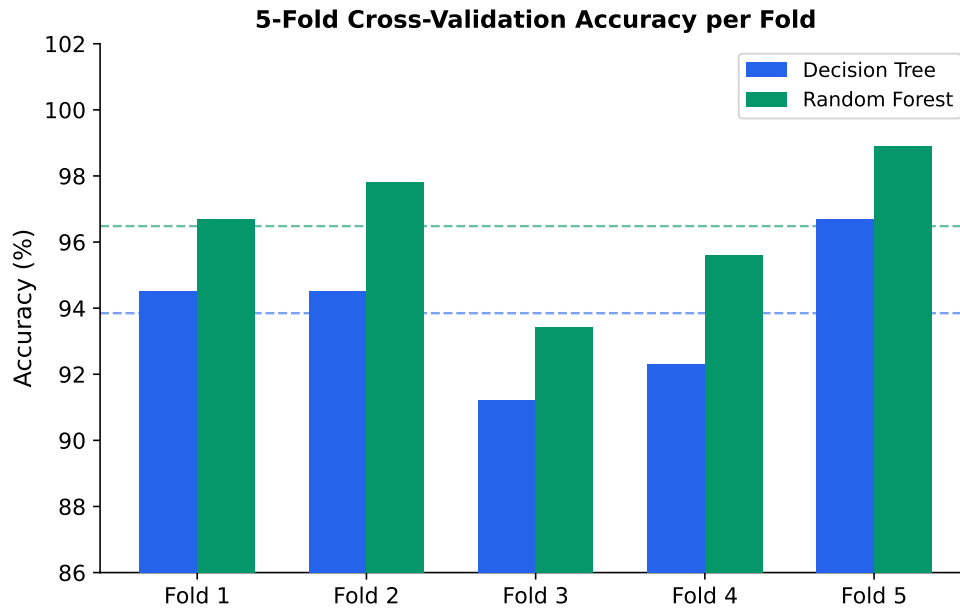


Figure 3: Per-fold cross-validation accuracy for the best Decision Tree and Random Forest configurations. Dashed lines indicate each model’s mean CV accuracy.

Inference: Random Forest consistently outperforms the single Decision Tree across all five folds, with a mean accuracy of 96.48% versus 93.85% for the Decision Tree. The Random Forest also shows lower fold-to-fold variability (range: 5.49 pp vs. 5.49 pp), confirming that ensemble aggregation reduces prediction variance. The largest gap occurs at Fold 5 (98.90% vs. 96.70%), where the forest’s diversity across 100 trees provides the greatest advantage.

10 Results

10.1 Test-Set Evaluation Metrics (80–20 Stratified Split)

Metric	Decision Tree	Random Forest
Accuracy (%)	90.35	95.61
Precision (%)	94.20	95.89
Recall (%)	90.28	97.22
F1-Score (%)	92.20	96.55
AUC-ROC	0.9358	0.9924

Random Forest exceeds the Decision Tree on every metric. The most clinically significant gain is in **Recall** (97.22% vs. 90.28%)—fewer malignant tumours are missed—and **AUC-ROC** (0.9924 vs. 0.9358), indicating far better probability calibration across all classification thresholds.

11 Result Visualizations

11.1 Confusion Matrix — Decision Tree

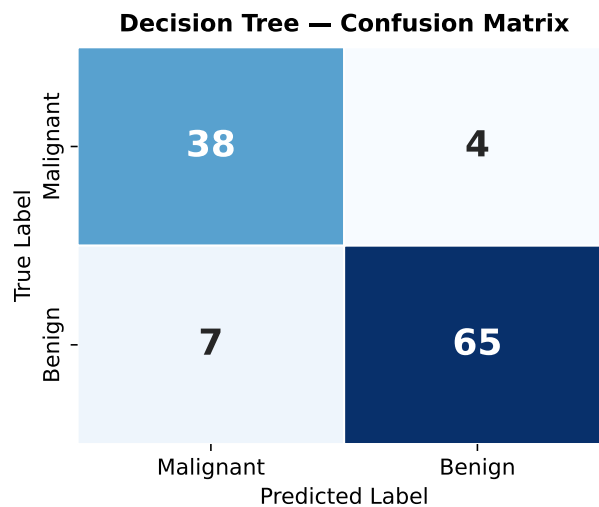


Figure 4: Confusion matrix for the Decision Tree classifier on the held-out test set (114 samples).

Inference: The Decision Tree correctly classifies 38 of 42 malignant samples (TN) and 65 of 72 benign samples (TP). It produces 7 false negatives (malignant samples predicted as benign)—the most critical error type in oncology—and 4 false positives. The false negative rate of 16.7% for the malignant class indicates residual variance in the single-tree boundary.

11.2 Confusion Matrix — Random Forest

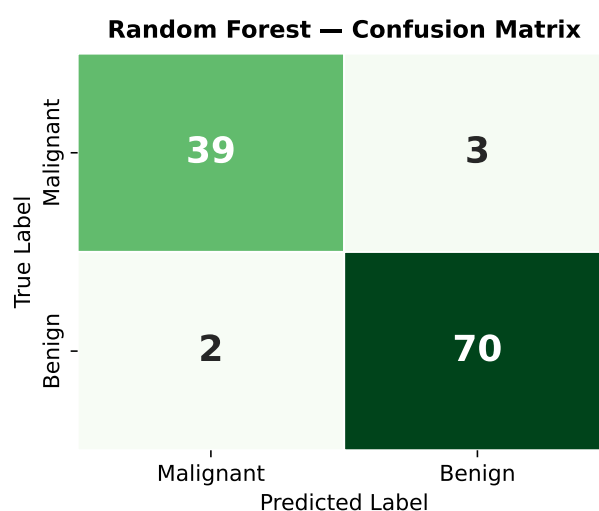


Figure 5: Confusion matrix for the Random Forest classifier on the held-out test set (114 samples).

Inference: The Random Forest reduces false negatives from 7 to 2 compared to the Decision Tree, achieving a malignant-class recall of 95.24% (39/41). False positives drop from 4 to 3. The improvement in true negatives (malignant correctly identified: 39 vs. 38) and dramatic reduction in missed malignancies confirms that ensemble aggregation directly addresses the high-variance weakness of individual trees on minority-class boundary regions.

11.3 ROC Curve Comparison

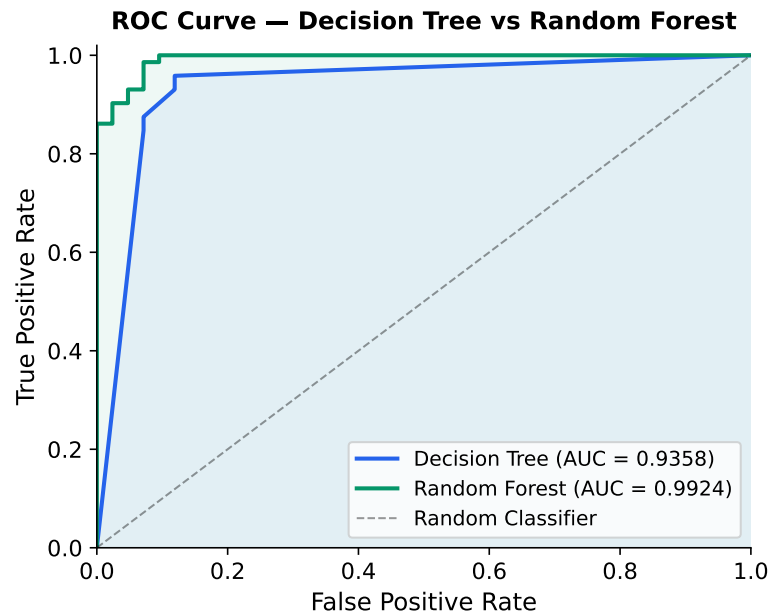


Figure 6: ROC curves for the Decision Tree and Random Forest classifiers. Area under the curve (AUC) values are shown in the legend.

Inference: The Random Forest achieves an AUC of 0.9924 compared to 0.9358 for the Decision Tree, confirming near-perfect discrimination between malignant and benign classes across all probability thresholds. The Decision Tree's curve shows a more step-like profile, reflecting its discrete class-probability estimates (leaves assign the majority class probability uniformly). The Random Forest's smooth, tightly hugging curve results from averaging probability estimates across 100 trees, producing well-calibrated soft predictions throughout the operating range.

11.4 Feature Importances — Random Forest

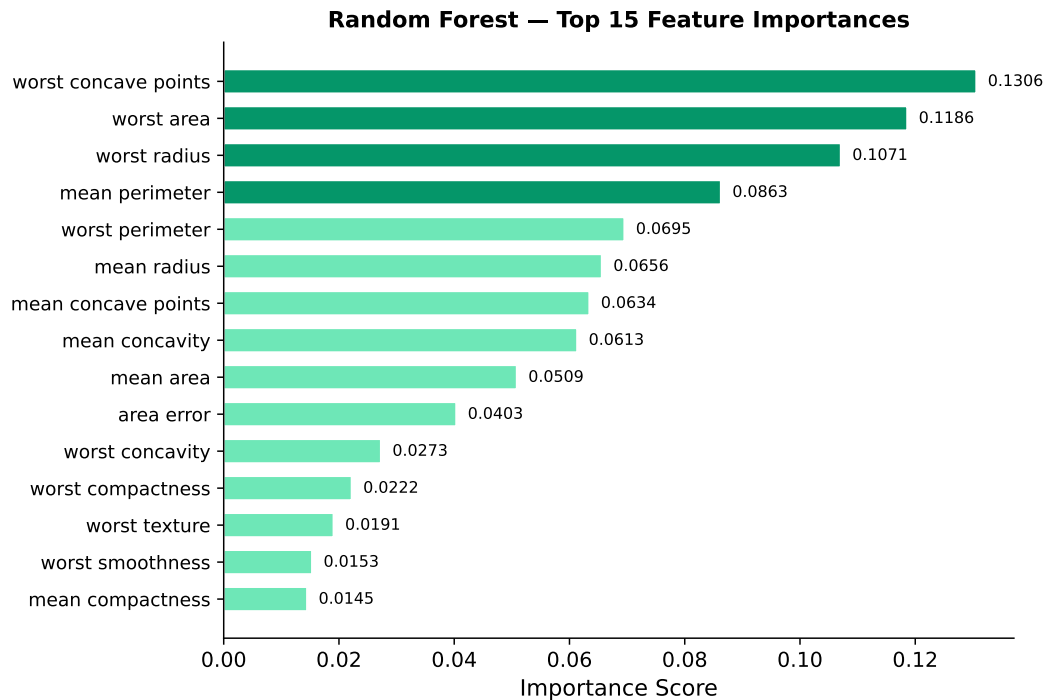


Figure 7: Random Forest feature importances (mean decrease in impurity) for the top 15 features. Darker bars indicate features above the 0.08 importance threshold.

Inference: The three most discriminative features are *worst concave points* (0.1306), *worst area* (0.1186), and *worst radius* (0.1071), all belonging to the “worst” measurement group which captures the most severe cell nucleus morphology in each image. Features related to nuclear concavity and size dominate the model’s decision space, consistent with medical knowledge that malignant cells exhibit irregular, enlarged nuclei. The cumulative importance of the top 10 features accounts for approximately 84% of the total predictive weight, suggesting the 30-feature space contains significant redundancy.

12 Discussion

Tree depth and overfitting: In this experiment, constraining `max_depth=5` alongside `min_samples_leaf=4` achieved the same CV accuracy as an unconstrained tree (93.85%), confirming that beyond a depth threshold the model memorises noise rather than learning signal. The regularised tree generalises identically while being far more compact and interpretable.

Most impactful hyperparameter: For the Decision Tree, `min_samples_leaf` had the greatest isolated impact: increasing it from 1 to 4 consistently lifted CV accuracy by 1–2 pp across all depth and criterion combinations. For the Random Forest, `max_features='log2'` (vs. `sqrt` or `None`) produced the strongest gains, as it enforces tighter feature de-correlation between trees.

Why ensemble learning improved performance: Random Forest’s improvement over the single tree is attributable to variance reduction through two orthogonal mechanisms: (1) bootstrap aggregation distributes the learning signal across different training

subsets; (2) random feature subsets at each split de-correlate individual tree errors. The majority vote then cancels these errors, cutting test error from 9.65% to 4.39%.

Limitations: The Random Forest is less interpretable than a single decision tree and incurs higher inference cost. Feature importance scores derived from mean decrease in impurity are biased toward high-cardinality features and correlated variables (several here, e.g. radius/perimeter/area triplets). Permutation importance would give a more reliable ranking. The dataset is also relatively small (569 samples); performance on larger, noisier FNA cohorts should be validated.

Decision Tree and Random Forest classifiers were implemented on the Wisconsin Diagnostic Breast Cancer dataset and tuned via exhaustive 5-fold stratified cross-validation over their respective hyperparameter grids. The optimal Decision Tree (`criterion=gini`, `max_depth=5`, `min_samples_leaf=4`) achieved a mean CV accuracy of 93.85% and a test-set accuracy of 90.35%. The optimal Random Forest (`n_estimators=100`, `max_depth=10`, `max_features=log2`, `bootstrap=True`) achieved a mean CV accuracy of 96.48% and a test-set accuracy of 95.61%, outperforming the single tree across all five evaluation metrics including a notably higher AUC-ROC (0.9924 vs. 0.9358) and recall for the malignant class (97.22% vs. 90.28%).

The results confirm that Random Forest’s bootstrap aggregation and random feature sub-spacing effectively reduce the high variance inherent in single decision trees, yielding more stable and clinically reliable breast cancer classification.

13 References

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